CareLoop SG

Quiet, AI-assisted support for seniors with high blood pressure, between doctor visits

THE PROBLEM

Most of hypertension care happens at home, between visits. Readings drift, pills get missed, a senior feels unwell and says nothing. Families worry from afar, and seniors who live alone may have no one checking at all. Alert apps make it worse: a fixed "BP over 140" alarm fires so often that families stop listening.

WHAT CARELOOP DOES

A short daily check-in, by voice or big buttons, in English, Singlish or Mandarin. CareLoop learns **each senior's own normal** and stays quiet while things are normal for them. When something changes, it steps up gently: it asks, then nudges, then notes it, and only hands over to the family (or to their Active Ageing Centre if they have no family) when a problem persists, with the evidence. **Red flags** (a dangerous reading or a danger symptom) reach the family at once, every time.

WHO USES IT

- **Seniors:** a large-text, low-effort app; it explains what each medicine is for and which vaccines or check-ups are due.
- **Family:** a quiet app with the trends, the evidence when needed, a shared care plan, and a one-page GP brief to print.
- **Centre staff:** a caseload console for seniors who live alone, sorted by who needs a call first.

KEY FEATURES

- Personal baselines and context (rain, haze, festivals) before anyone is alerted
- Handovers that wait for a person to choose a step; CareLoop never calls anyone itself
- Consent: the senior chooses what family sees
- Readings from a Fitbit or OMRON cuff via Apple Health / Health Connect (simulated in the demo)
- Medicine notes in plain words; an unknown medicine goes to the pharmacist, never guessed

HOW IT USES AI, SAFELY

Rules decide; the AI only reads and writes words. A large language model reads each check-in to catch what simple rules miss (in a live test, "my chest feel like got elephant sitting on top" became a chest-pain red flag that keyword rules alone had missed), and words the gentler messages in the senior's own language and tone. It never chooses who is told or whether a number is dangerous. Every message passes a guardrail (no diagnosis, no dose changes, no invented numbers) or a fixed template goes out; red flags always use fixed wording. With no model at all, CareLoop still works on rules and templates.

TESTED BLIND

On a frozen, held-out set of synthetic seniors (720 days), compared with the usual "alert above 140" rule:

5 of 5	**16 of 17**	**0**
red flags caught (fixed rule: 3 of 5)	real problems handed over	false alarms per senior-month (fixed rule: 1.29)

The honest trade-off: a slow blood-pressure drift is handed over about 3 days later than the fixed rule. In return, families get no false alarms, and CareLoop sees what one number can't: missed medicine, days unwell, isolation.

HOW IT'S BUILT

React web app (installable on a phone) · Python FastAPI backend · a rule engine with one thresholds file · a language layer that is the only path to the model (Tencent Hunyuan or any OpenAI-compatible model) · open data from data.gov.sg (haze, rainfall, centre locations) · 183 offline tests.

Live demo: careloop-sg.onrender.com (may take a minute to wake) · **Senior app:** /senior?senior=tan · **Family:** /care/family · **Centre:** /care/aac
Code: github.com/DrunkWiz/careloop-sg · **Architecture:** docs/architecture.md · **Evaluation:** docs/evaluation.md

All people in the demo are synthetic. CareLoop is a student prototype, not a medical device, and is not connected to any clinic system. Clinical thresholds, red-flag wording and medicine notes are placeholders until reviewed by a clinician. Team: Nathan, Mingyu, Vibu.